KAILA UYEDA

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EDUCATION

UC Irvine *2024-*
Ph.D. in Earth System Science

Haverford College *2019-2023*
B.S. in Mathematics; Minor in Health Studies
Thesis Title: *Analysis and Modification of a Sinking Model for Biofouled Particles in the Ocean*

EMPLOYMENT/RESEARCH EXPERIENCE

Graduate Student Researcher, UC Irvine *2024-*
Advised by Henri Drake (UC Irvine)

Parallel Computing Summer Intern, Los Alamos National Laboratory *2025*
Advised by Mark Petersen, Katherine Smith, Dylan Schlichting (Los Alamos National Laboratory)

Research Assistant in Physical Oceanography, Texas A&M University *2023-2024*
Advised by C. Spencer Jones (Texas A&M University)

Earth Science on Volcanic Islands REU Summer Intern, UH-Mānoa *2022*
Mentored by James Potemra (UH-Mānoa)

Chemistry Research Assistant, Haverford College *2022*
Advised by Helen White (Haverford College)

Mathematics Independent Study, Haverford College *2021*
Advised by Amanda Laubmeier (Texas Tech Univesity) & Rebecca Everett (Haverford College)

AWARDS

LANL Center for Space and Earth Science Student Fellowship *2026-2028*

GAANN Fellowship *Spring 2026*

UCI ESS Outstanding Graduate TA Award *Winter 2026*

RESEARCH PUBLICATIONS

Submitted / Under Review

Drake, H., Bonan, D., Keshwani, I., Liu, R., Meza, A., Poinelli, M., Rafter, P.A., Rugenstein, M., Uyeda, K., Willcott, K. Collapse and recovery of abyssal circulation and ventilation under warming. *Under review at Nature Geoscience.*

Satish, S., Uyeda, K., Jones, C.S. Impact of Equatorial Wind Change on the Meridional Heat Transport in the Atlantic. *Submitted.* https://doi.org/10.31223/X5QJ1T

Published / In Press

Kyle E. Hinson, Dylan R. Schlichting, Robert D. Hetland, Darren Engwirda, Katherine M. Smith, Mark R. Petersen, **Kaila Uyeda** (Oct. 2026). "Towards high-fidelity simulations of coastal submesoscale baroclinic instabilities with MPAS-O (vE3SM3.0.0) – Part 1: Idealized experiments". English. In: *Geoscientific Model Development* 19.21, pp. 10321–10344. DOI: 10.5194/gmd-19-10321-2026.

TEACHING

EarthSS 40C - Earth System Physics. Teaching Assistant. *Fall 2026*

EarthSS 053 - The Ocean System. Teaching Assistant. *Winter 2026*

SELECTED PRESENTATIONS

Uyeda, K., Schlichting, D., Smith, K., Drake, H. (2026) **Convergence testing two numerical mixing diagnostics in idealized and realistic E3SM simulations.** *Coastal, Ocean, Sea Ice Modeling Seminar*, Los Alamos, NM [TALK]

Uyeda, K., Smith, K., Schlichting, D., Drake, H. (2026). **Testing the convergence of two numerical mixing diagnostics on an unstructured grid.** *Center for Space and Earth Science Symposium*, Los Alamos, NM. [TALK]

Uyeda, K., Schlichting, D., Smith, K., Hetland, R., Petersen, M., Drake, H. (2026). **The impacts of horizontal resolution in simulations of submesoscale baroclinic instabilities in MPAS-Ocean.** *Ocean Mixing Gordon Research Conference*, Mount Holyoke, MA. [POSTER]

Uyeda, K., Undzis, B., Schlichting, D., Smith, K., Petersen, M. (2025). **Quantification of numerical mixing in ocean models.** *Parallel Computing Summer Research Internship Symposium*, Los Alamos, NM. [TALK]

Uyeda, K., Jones, C.S. (2024). **Understanding the impact of vertical diffusivity on northward water pathways through the tropical Atlantic.** *American Meteorological Society 24th Conference on Atmospheric and Oceanic Fluid Dynamics*, Burlington, VT. [POSTER]

Uyeda, K., Potemra, J. (2022). **Long-term trajectories of subsurface ocean debris in the North Pacific.** *UH-Mānoa SURE Symposium*, Honolulu, HI. & *American Geophysical Union Fall Meeting 2022*, Chicago, IL. [POSTER]

Uyeda, K., Youngs, S., Anderson, C., Grzenda, D., Morrison, A., White, H., Michel, A. (2022). **Examining the relationship between weathering and the accumulation of metals on macroplastic debris collected from coastal environments.** *Ocean Sciences Meeting (Virtual)*, Honolulu, HI. [POSTER]

Uyeda, K., Laubmeier, A., Everett, R. (2022). **Modeling food web interactions based on predator movements and habitat use.** *Nebraska Conference for Undergraduate Women in Mathematics (Virtual)*, University of Nebraska. [POSTER]

FIELD EXPERIENCE

Lewes Beach, Cape Henlopen State Park, Prime Hook National Wildlife Park (10/02/21) Collection of oil residues from October 2020 mystery oil spill in the Delaware Bay.

Pirates Cove Beach, Newport Beach (10/24/26, 11/07/26) Coastal Observations and Volunteer Ecology tidepool monitoring.

LEADERSHIP & SERVICE

Departmental Leadership

Graduate Student Representative, **UCI Earth System Science** *2026-2027*

Co-President, **Society for Women in Marine Science at UCI** *2025-*